



Exploration of treatment strategies and susceptibility gene of postoperative nausea and vomiting in breast cancer patients: a randomised controlled trial

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Abstract

Background A history of severe nausea and vomiting during pregnancy (SNVP) is a risk factor for postoperative nausea and vomiting (PONV). This study aimed to explore potentially effective treatment strategies and potential genetic factors underlying SNVP risk-related PONV.

Methods A total of 140 female patients undergoing breast cancer surgery were assigned to either the study group (70 with SNVP) or the control group (70 with mild to moderate nausea and vomiting during pregnancy (MNVP)). Patients in each group were randomly assigned to two different treatment subgroups and received either ondansetron plus dexamethasone (OD) or OD + TEAS (ODT) (transcutaneous electrical acupoint stimulation, TEAS). Blood samples were collected from patients before induction (D0) and 24 h (D1) after surgery for growth differentiation factor 15 (GDF-15) evaluation. The primary outcome was the incidence of PONV within 36 h. The secondary outcome was the serum GDF-15 level.

Results The incidence of PONV in the SNVP group was significantly higher than that in the MNVP group within 24 h ($P < 0.005$). In the SNVP group, ODT-treated patients had less PONV than those in the OD-treated group during the 6–12 h ($P = 0.033$) and 12–24 h ($P = 0.008$) intervals, while within 6 h, there were fewer vomiting cases in the ODT-treated group (SNVP-ODT vs. SNVP-OD, 7/33 vs. 19/35, $P = 0.005$). The preoperative GDF-15 serum levels in patients with SNVP were significantly higher ($P = 0.004$). Moreover, higher preoperative GDF-15 serum levels correlated with a higher incidence of PONV ($P = 0.043$).

Conclusions TEAS showed significant effect on PONV treatment in patients with SNVP. A higher serum GDF-15 level was associated with a history of SNVP, as well as a higher risk of PONV.

Keywords Breast cancer surgery · Transcutaneous electrical acupoint stimulation · Nausea and vomiting during pregnancy · Postoperative nausea and vomiting · Growth differentiation factor 15

Introduction

Patients undergoing breast cancer surgery under general anesthesia experience a common adverse phenomenon called postoperative nausea and vomiting (PONV) [1]. PONV can lead to prolonged hospital stay, wound dehiscence, and delayed nutrition, and is associated with clinical and economic consequences [2, 3]. It is widely known that the main risk factors for PONV are female sex, non-smoking status, history of motion sickness or PONV, and the use of postoperative opioids [4]. Patients with breast cancer usually face at least two risk factors: female sex and non-smoking, which is a high-risk population for PONV. However, according to the results of one of our previous studies [5], severe nausea and vomiting during

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pregnancy (SNVP) is another risk factor for PONV in breast cancer patients. Although ondansetron combined with dexamethasone is an effective routine prophylactic plan, it worked poorly in patients with SNVP, with approximately 50% experiencing PONV and more than 70% of the PONV patients being severe cases [5]. Based on this background, we aimed to explore potential effective treatment strategies and possible causes of PONV in those SNVP-risk patients. Additionally, the association between SNVP and PONV was reconfirmed.

Transcutaneous electrical acupoint stimulation (TEAS) is a form of acupuncture that applies electrical stimulation at specific acupoints through cutaneous electrodes. TEAS is a non-pharmacological and non-invasive treatment method that has been used in perioperative anesthesia management to improve patient recovery, including PONV prevention [6–8] and postoperative pain reduction [9]. In this study, we evaluated whether TEAS, along with ondansetron and dexamethasone, could decrease the risk of PONV in patients with SNVP. According to the theory of traditional Chinese medicine, bilateral Hegu (LI4) and Neiguan (PC6) were chosen as acupuncture points, which have been frequently recommended for PONV prevention [10–12].

PONV is a multifactorial condition. Hereditary factors might play a crucial role in predisposing patients with SNVP risk to PONV. Although hereditary factors associated with the development of PONV or nausea and vomiting during pregnancy (NVP) have been frequently discussed [13, 14], there was no identical genetic element has been reported to be associated with either PONV or NVP. In 2018, Fejzo et al. [15] published a genome-wide association study that identified growth differentiation factor 15 (GDF-15) as a possible genetic factor influencing SNVP. Another interesting study by Fejzo et al. [16], which was published in *Nature*, affirmed that GDF15 is linked to maternal risk of nausea and vomiting during pregnancy. GDF15 triggers anorexia through the induction of nausea and/or emetic neurocircuitry [17]. GDF15 signaling has also been implicated in the development of NVP [18]. Accordingly, we hypothesized that GDF-15 might be an identical genetic element associated with both PONV and NVP, predisposing patients with SNVP to the risk of PONV. In summary, this study reconfirmed the association between SNVP and PONV, explored treatment strategies of SNVP-risk patients, and evaluated serum levels to determine whether GDF-15 is a genetic factor that increases the development of PONV in SNVP-risk patients.

Methods

This prospective trial was performed between February and August 2022, and prospectively registered in Chictr.org.cn registry system on 28 November 2021

(ChiCTR2100053752). Ethical approval for this study (approval number: 21/391-3062) was provided by the Ethical Committee of National Cancer Center/Cancer Hospital, Beijing, China on 9 September 2021. This work has been reported in line with the Consolidated Standards of Reporting Trials (CONSORT) Guidelines (www.consort-statement.org) [19].

Study participants

Seventy sequential female patients with a history of SNVP, who were planning to undergo breast cancer surgery, were enrolled in this prospective study. At the same time, patients with mild-moderate nausea and vomiting during pregnancy (MNVP) (who were operated on by the same surgeons) were chosen as controls (SNVP group vs. MNVP group). Written informed consent was obtained from all enrolled patients. The inclusion criteria were as follows: (1) age 18–60 years, (2) history of at least one full-term pregnancy, (3) American Society of Anesthesiologists physical status I or II, and (4) unilateral breast cancer confirmed by histopathology or cytopathology before surgery. Exclusion criteria included a history of motion sickness, diabetes, gastrointestinal diseases, peptic ulcers, endocrinological disorders, liver diseases, use of corticosteroids or antiemetic medications within 24 h before surgery, a history of allergy to any medication used in the study protocol, active drug abuse, pregnancy or breastfeeding, and uncertainty of the severity of NVP.

An investigator collected the patients' history of SNVP before surgery. For the classification of the severity of NVP in participants, we referred to the method used by Zhang [20] in a study that found familial aggregation of SNVP. This study defined the degree of NVP based on the methods used to determine the severity of NVP in the participants' family members. Patients were asked to report the severity of NVP according to the following definitions [20]: (1) No NVP-no nausea or vomiting. (2) Very little NVP-felt nauseous and/or vomited for less than seven days. (3) Typical NVP-nausea and vomiting lasting more than seven days; able to sustain a normal daily routine; without weight loss; no need to consult a health professional. (4) More severe NVP-persistent nausea and vomiting that interfered with normal daily routine may have consulted a health professional and may have lost a few pounds. (5) Hyperemesis Gravidarum-persistent nausea and vomiting with weight loss that significantly interfered with daily routine; medical treatments, such as medications or nutritional therapy, were needed to prevent weight loss and/or nausea/vomiting. In this study, patients with items 1–3 were regarded as having mild-moderate NVP (MNVP), while 4–5 were defined as having SNVP.

Randomization

Seventy patients from each group were then randomly allocated into the ondansetron plus dexamethasone (Ondan + Dex, OD) or Ondan + Dex + TEAS (ODT) treatment subgroups (SNVP-OD group vs. SNVP-ODT group, MNVP-OD group vs. MNVP-ODT group). Prior to study initiation, 140 sequentially numbered envelopes containing the allocations were prepared. A physician independent of the study randomly inserted 70 treatment strategies for each group into the envelope. A random allocation sequence was generated by using computer-generated random numbers. The researchers opened an envelope to determine the treatment strategy before the induction of general anesthesia. This was a double-blinded study. The investigator who administered the interventions was not blinded. Outcomes were assessed by trained investigators who were blinded to the patients' allocation. Patients and other clinical staff, including anesthesiologists, surgeons, and ward staff, were unaware of this allocation.

Intervention

Standard general anesthesia was induced with propofol 1–2 mg kg⁻¹, sufentanil 0.3–0.6 µg kg⁻¹, and cis-atracurium 0.2–0.4 mg kg⁻¹. Additionally, 5 mg of dexamethasone was used as antiemetic prophylaxis in all enrolled patients [21]. After inserting a laryngeal mask airway, the patients were mechanically ventilated to maintain the end-tidal carbon dioxide concentration at 35–45 mmHg, with a fresh gas flow of 2 L/min 60% oxygen. 1.5–2.5% sevoflurane was constantly inhaled for maintenance of anesthesia. Sufentanil 0.1 µg kg⁻¹ was added intraoperatively, as required. At the end of the surgery, 100 mg flurbiprofen axetil and 4 mg ondansetron were administered to all patients. After the operation, all patients were extubated and transferred to the post-anesthesia care unit.

Before anesthetic induction, a HANS-200A Acupoint Stimulator (Nanjing Jisheng Company, China) was used to stimulate the bilateral Neiguan (PC6) and Hegu (LI4) acupoints for 30 min. The LI4 acupoint is positioned between the first and second metacarpal bones at the highest point of prominence when the thumb and index finger are adducted [22]. With the forearm supine, the PC6 acupoint is located between the flexor carpi radialis muscle and palmaris longus tendon, 2 cm proximal to the midpoint of the volar transverse carpal crease [22]. The frequency of the TEAS device was set at 2/100 Hz. In the TEAS-treated group, an intensity of 10 mA was administered. Meanwhile, Participants in the control group received sham stimulation at the same points using sham electrode wires, cutting the output wires and resulting in no current output being delivered to the patient despite being connected. The researchers also informed each

patient that they might not have felt a tingling sensation when the intervention was applied. Additionally, to make the simulation as realistic as possible, the stimulator with the dummy electrode wires had the same operating power indicator and sound as the normal electrode wires, indicating that the stimulator was operating effectively even though there was actually no current output.

Outcomes

The primary outcome was the incidence of PONV within 36 h. Episodes of vomiting and severity of nausea were recorded at 2, 6, 12, 24, and 36 h after the operation. The degree of nausea was measured using a visual analog scale (VAS; 0 = no nausea to 10 = most severe nausea) [23]. A score of < 5 was considered mild nausea and ≥ 5 was considered severe nausea. Vomiting episodes ≥ 2 were considered severe and those < 2 were considered mild. For severe PONV, 4 mg ondansetron was administered intravenously. Postoperative pain was assessed simultaneously using VAS (0 = no pain to 10 = most severe pain). If the pain VAS score was ≥ 3, 100 mg flurbiprofen axetil was administered.

The secondary outcome was the serum GDF-15 level. GDF15's short half-life makes it possible to track changes in serum levels within hours [24]. Most PONV occurs within 24 h after the operation. Therefore, we chose to detect the level of GDF-15 at the 24 h postoperative point to analyze the association between GDF-15 levels and the incidence of PONV. Blood samples were collected from the patients and stored in heparinized tubes before induction (D0) and 24 h (D1) after surgery in the ward. The samples were centrifuged for 20 min after collection, and the extracted plasma was frozen in liquid nitrogen and stored at -80 °C until analysis. Plasma GDF-15 concentrations were measured using ELISA (Human GDF-15 ELISA Kit, Multisciences China) with a linear range from 15 to 1000 pg ml⁻¹.

Statistical analysis

Sample sizes were calculated based on our preliminary experiment. We enrolled 11 and 13 patients in the SNVP-OD and SNVP-ODT groups, respectively. The incidence rates of PONV in the two groups were 54.5% and 23.1%, respectively. Therefore, it was determined that in order to detect a significant difference with a power of 0.8 at α = 0.05, there would need to be at least 34 patients in each group. 35 patients in each treatment subgroup were recruited to ensure a sufficiently large sample size. The SPSS 23.0 for windows (SPSS, Inc., Chicago, IL, USA) was used for data analysis. Normally distributed continuous data are expressed as mean (SD) and were analyzed by analysis of variance (ANOVA), independent-sample t-test, or paired t-test. Non-normally distributed variables are presented as medians (interquartile

ranges) or numbers/percentage of patients. Nonparametric data were analyzed using Mann–Whitney and Wilcoxon tests. Two-sided tests were used to determine the statistical significance at $P < 0.05$.

Results

Baseline characteristics of the patients

Patient enrollment started on February 1st, 2022. A total of 138 patients were included in the study (Fig. 1). Demographic characteristics, including age, body weight, height, body mass index (BMI), smoking status, and history of PONV, are shown in Table 1. There were no differences

Fig. 1 Flow of patients throughout study

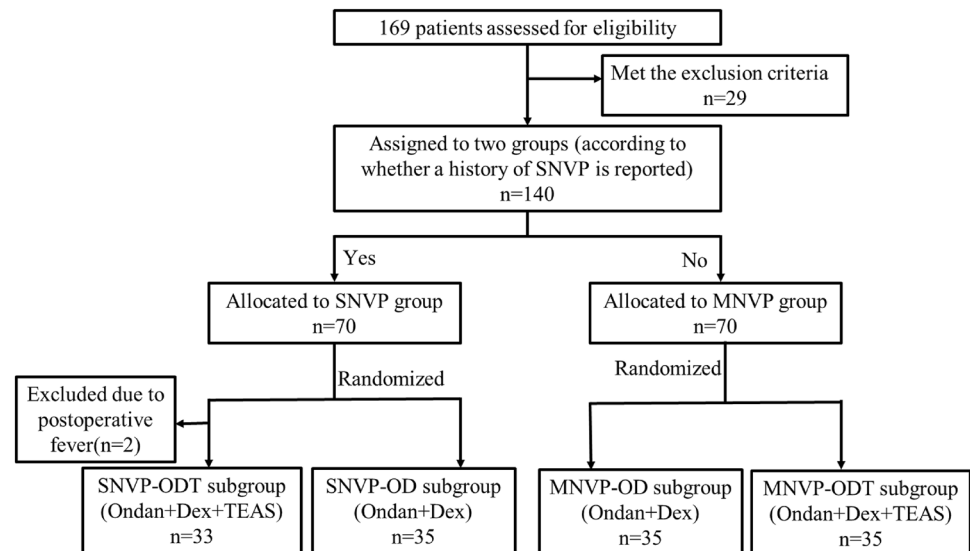


Table 1 The baseline characteristics of patients

	SNVP-OD group (n = 35)	SNVP-ODT group (n = 33)	<i>P</i>	MNVP-OD group (n = 35)	MNVP-ODT group (n = 35)	<i>P</i>
Age (years) ^a	49.9 ± 9.5	47.4 ± 8.2	0.258	49.3 ± 10.0	49.0 ± 8.5	0.888
Weight (kg) ^a	63.7 ± 9.5	61.0 ± 7.4	0.21	64.2 ± 11.7	62.7 ± 10.6	0.58
Height (cm) ^a	160.7 ± 4.1	160.6 ± 4.3	0.848	161.6 ± 5.4	160.2 ± 6.2	0.307
Body mass index ^a	24.6 ± 3.6	23.7 ± 2.7	0.225	24.5 ± 4.0	24.4 ± 3.6	0.874
Smoking (%)	2(5.7%)	1(3.0%)	0.59	1(2.9%)	2(5.7%)	0.555
History of PONV (%)	1(2.9%)	1(3.0%)	0.966	2(5.7%)	3(8.6%)	0.643
Neo-adjuvant chemotherapy (%)	15(42.9%)	11(33.3%)	0.419	9(25.7%)	9(25.7%)	1.00
Surgical procedure						
Radical mastectomy + axillary dissection (%)	25(71.4%)	12(36.4%)	0.004	14(40.0%)	8(22.9%)	0.122
Radical mastectomy + SNLB (%)	3(8.6%)	7(21.2%)	0.141	13(37.1%)	11(31.4%)	0.615
Segmental mastectomy + SNLB (%)	6(17.1%)	9(27.3%)	0.314	6(17.1%)	5(14.3%)	0.743
Segmental mastectomy + axillary dissection (%)	1(2.9%)	4(12.1%)	0.144	1(2.9%)	7(20.0%)	0.024
Partial mastectomy (%)	0 (0%)	1(3.0%)	0.299	1(2.9%)	4(11.4%)	0.164
Surgery time (min) ^a	96.2 ± 28.3	106.6 ± 38.0	0.204	84.9 ± 28.0	84.9 ± 29.7	0.99
Length of anesthesia (min) ^a	120.9 ± 31.8	135.1 ± 44.5	0.132	105.2 ± 31.0	105.2 ± 31.5	0.997
Sufentanil (μg) ^a	21.9 ± 3.2	21.2 ± 3.1	0.402	20.9 ± 3.1	21.0 ± 3.4	0.854

PONV postoperative nausea and vomiting, SNLB sentinel lymph node biopsy

^aValues are expressed as the mean ± standard deviation

in the duration of anesthesia or surgery. The number of patients who received neoadjuvant chemotherapy in the OD and ODT treatment subgroups was similar, and most of the different surgical procedures were well balanced between the different treatment subgroups (SNVP-OD vs. SNVP-ODT, and MNVP-OD vs. MNVP-ODT) (Table 1). However, due to unpredictable sentinel lymph node involvement, more patients in the SNVP-OD group underwent total mastectomy + axillary dissection than those in the SNVP-ODT group ($P=0.004$; Table 1), while more patients in the MNVP-ODT group underwent segmental mastectomy + axillary dissection than those in the MNVP-OD group ($P=0.024$; Table 1).

The incidence of PONV in different groups

The incidence of PONV in the SNVP group was significantly higher than that in the MNVP group at 2, 6, 12, and 24 h, with no significant differences at 36 h after the operation (Table 2). Overall, the incidence of PONV was lower in the MNVP group than in the SNVP group (Table 2). Severe PONV (nausea score ≥ 5 and/or vomiting episode ≥ 2) occurred more often in the SNVP group than in the MNVP group (25/68 vs. 3/70, $P < 0.001$). Among the severe cases, in the SNVP group, 14 occurred within 6 h after the operation, while 7 within 2 h, 3 within 12 h, and 1 within 24 h (in the MNVP group, 1 within 2 h, 1 within 6 h, and 1 within 12 h). In the subgroup

analysis, ODT-treated patients in the SNVP group (SNVP-ODT group) had less PONV than the patients in the SNVP-OD group during the 6–12 h ($P=0.033$) and 12–24 h ($P=0.008$) intervals after the operation (Table 3), whereas less PONV occurred in the patients in the MNVP-ODT group than in the MNVP-OD group during the 0–2 h ($P=0.022$) and 2–6 h ($P=0.04$) intervals after the operation (Table 3). Although no significant difference in PONV incidence was observed during 0–6 h intervals between the OD- and ODT-treated groups in the SNVP group, there were fewer vomiting cases in the SNVP-ODT group than in the SNVP-OD group (SNVP-ODT vs. SNVP-OD, 7/33 vs. 19/35, $P=0.005$).

Association of GDF-15 with the history of SNVP and the incidence of PONV

The postoperative mean serum levels of GDF-15 on Day 1 were significantly higher than those on day 0 in all the patients (Fig. 2, $P < 0.001$). The preoperative serum concentrations of GDF-15 in patients receiving total mastectomy plus axillary dissection were significantly higher than those in patients receiving other surgical procedures (total mastectomy plus axillary dissection vs. other surgical procedures, 153.14 ± 63.60 vs. 106.40 ± 44.76 , $P < 0.001$), suggesting that higher GDF-15 levels might predict more aggressive surgical procedures. Moreover, it was found that the preoperative serum concentrations of GDF-15 in patients receiving neo-adjuvant chemotherapy were significantly higher than in those who did not receive the therapy (with neo-adjuvant chemotherapy vs. without neo-adjuvant chemotherapy, 148.39 ± 70.06 vs. 116.08 ± 48.89 , $P=0.002$). Furthermore, the preoperative GDF-15 serum levels in patients with SNVP were significantly higher than those in patients with MNVP (Fig. 3a, $P=0.004$). At the same time, a higher preoperative GDF-15 serum level was also associated with a higher incidence of PONV (Fig. 3b, $P=0.043$). The baseline GDF-15 serum levels in patients with SNVP-OD and SNVP-ODT, or with MNVP-OD and MNVP-ODT were all similar (Fig. 3c, $P > 0.05$).

Discussion

Consistent with a previous study [5], this study confirmed that a history of SNVP in patients led to a higher incidence of severe PONV. In-depth exploration of treatment methods revealed that TEAS significantly improved the therapeutic effect within 24 h after the operation in patients with SNVP risk, in which the incidence of PONV was significantly reduced during 6–24 h, and the vomiting symptom was effectively controlled during 0–6 h by the ODT. A slightly positive effect was observed in the MNVP group. Analysis

Table 2 Incidence of PONV in the SNVP and MNVP groups

	SNVP Group (n = 68)	MNVP Group (n = 70)	P
0–2 h			
PONV	33 (48.5%)	11 (15.7%)	0.000
Asymptomatic	35 (51.5%)	59 (84.3%)	
2–6 h			
PONV	31 (45.6%)	10 (14.3%)	0.000
Asymptomatic	37 (54.4%)	60 (85.7%)	
6–12 h			
PONV	23 (33.8%)	5 (7.1%)	0.000
Asymptomatic	45 (66.2%)	65 (92.9%)	
12–24 h			
PONV	13 (19.1%)	2 (2.9%)	0.002
Asymptomatic	55 (80.9%)	68 (97.1%)	
24–36 h			
PONV	3 (4.4%)	0 (0%)	0.076
Asymptomatic	65 (95.6%)	70 (100%)	
Total			
PONV	34 (50.0%)	15 (21.4%)	0.000
Asymptomatic	34 (50.0%)	55 (78.6%)	

PONV postoperative nausea and vomiting

Table 3 Incidence of PONV in four subgroups

	SNVP-OD group (n = 35)	SNVP-ODT group (n = 33)	<i>P</i>	MNVP-OD group (n = 35)	MNVP-ODT group (n = 35)	<i>P</i>
0–2 h						
PONV	20 (57.1%)	13 (39.4%)	0.14	9 (25.7%)	2 (5.7%)	0.022
Asymptomatic	15 (42.9%)	20 (60.6%)	3	26 (74.3%)	33 (94.3%)	
2–6 h						
PONV	19 (54.3%)	12 (36.4%)	0.13	8 (22.9%)	2 (5.7%)	0.04
Asymptomatic	16 (45.7%)	21 (63.6%)	8	27 (77.1%)	33 (94.3%)	
6–12 h						
PONV	16 (45.7%)	7 (21.2%)	0.03	3 (8.6%)	2 (5.7%)	0.643
Asymptomatic	19 (54.3%)	26 (78.8%)	3	32 (91.4%)	33 (94.3%)	
12–24 h						
PONV	11 (31.4%)	2 (6.1%)	0.00	0 (0%)	2 (5.7%)	0.151
Asymptomatic	24 (68.6%)	31 (93.9%)	8	35 (100%)	33 (5.7%)	
24–36 h						
PONV	3 (8.6%)	0 (0%)	0.08	0 (0%)	0 (0%)	1.00
Asymptomatic	32 (91.4%)	33 (100%)	5	35 (100%)	35 (100%)	
Total						
PONV	20 (57.1%)	14 (42.4%)	0.22	10 (28.6%)	5 (14.3%)	0.145
Asymptomatic	15 (42.9%)	19 (57.6%)	5	25 (71.4%)	30 (85.7%)	

PONV postoperative nausea and vomiting

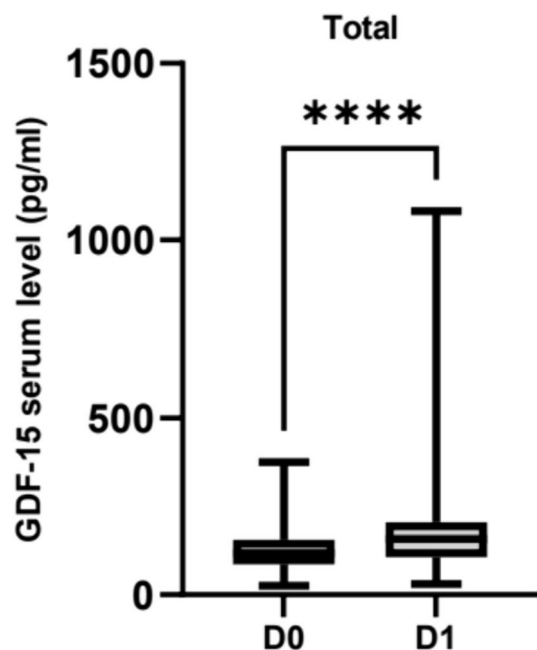


Fig. 2 Mean preoperative and postoperative serum GDF-15 concentrations in all the enrolled patients (n=138): the serum concentrations of GDF-15 in patients were significantly increased after surgeries (the preoperative GDF-15 (D0) vs. the postoperative GDF-15 (D1), $P < 0.001$)

of GDF-15 serum concentrations revealed that a higher level in the preoperative stage was significantly associated with the development of SNVP and a higher incidence of PONV.

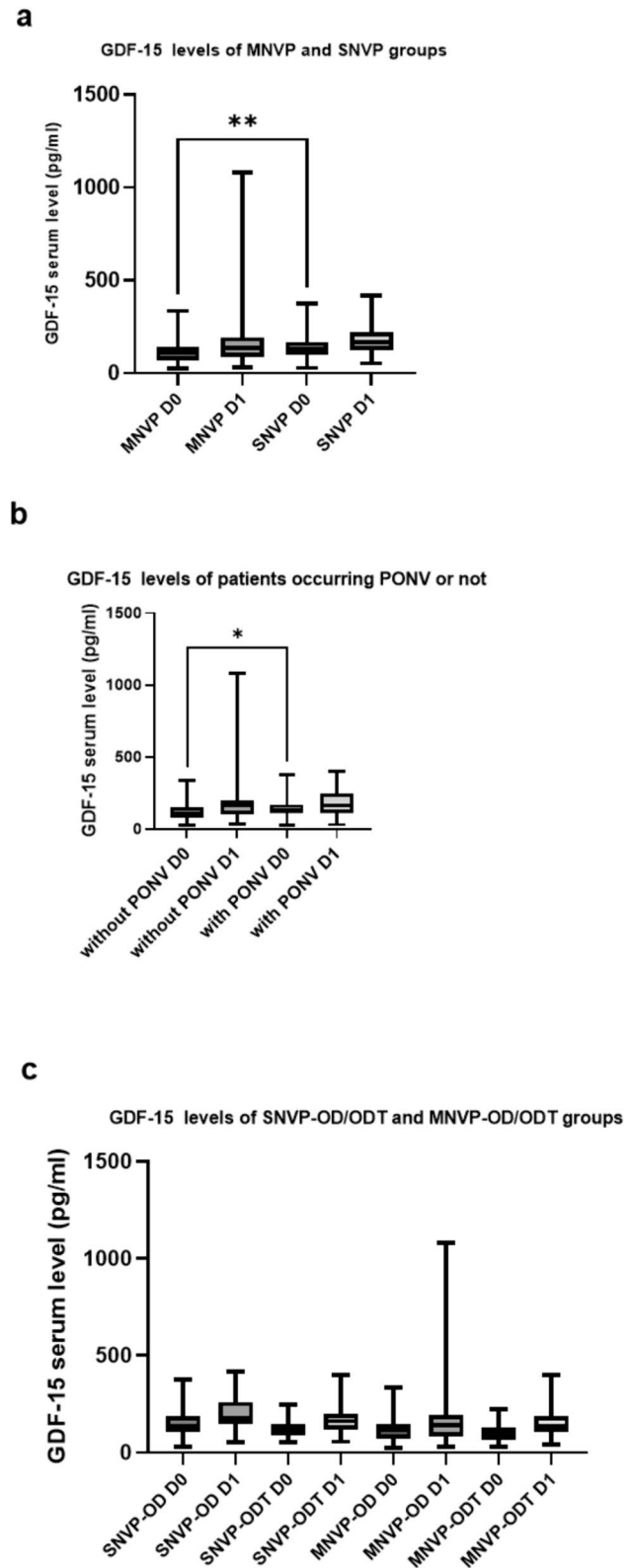
Our previous study showed that a history of SNVP may be a risk factor for PONV, with patients with a history of SNVP experiencing a 30.3% higher incidence of PONV and 31.6% more severe PONV [5]. Findings from this study corroborated our previous findings, with PONV occurring in 50% of patients in the SNVP group, which was 28.6% higher than that in the MNVP group. At the same time, the incidence of severe PONV in the SNVP and MNVP groups was 36.8% and 4.3%, respectively. These results highlight how a history of SNVP strongly predicts an increased risk for PONV, warranting the inclusion of SNVP in female patients as one of the risk factors for PONV prediction. More effective prophylactic agents should be recommended for female patients with a history of SNVP. Furthermore, the underlying etiology needs further investigation to identify better targeted treatments for application.

According to previous studies [5], the combination of dexamethasone and a 5-HT₃ receptor antagonist failed to effectively manage PONV after breast cancer surgery. More than 50% of the patients in the SNVP group still developed PONV. This study evaluated the potential role of TEAS in reducing the incidence of PONV in patients undergoing breast cancer surgery to identify the most effective treatment strategy. As shown in Table 3, the TEAS-combined strategy reduced the incidence of PONV at 6–12 h and 12–24 h intervals in the SNVP group, whereas in the MNVP group, its significant effect was during the 0–2 h and 2–6 h intervals. Although no significant difference was found in the incidence of PONV

Fig. 3 Mean preoperative or postoperative serum GDF-15 concentrations in patients with SNVP or MNVP history, in patients occurring PONV or not, and in patients in OD and ODT subgroups: **a** The preoperative (D0) serum concentrations of GDF-15 in patients with SNVP ($n=68$) were significantly higher than in those with MNVP ($n=70$) ($P=0.004$), and the postoperative (D1) serum concentrations of GDF-15 in patients with SNVP or MNVP increased significantly compared to D0 ($P<0.001$). **b** The preoperative (D0) serum concentrations of GDF-15 in patients occurring PONV (with PONV, $n=49$) were significantly higher than in those not (without PONV, $n=89$) ($P=0.043$), and the postoperative (D1) serum concentrations of GDF-15 in patients occurring PONV or not increased significantly compared to D0 ($P<0.001$). **c** The preoperative (D0) serum concentrations of GDF-15 in patients with SNVP-OD ($n=35$) and SNVP-ODT ($n=33$) were similar ($P>0.05$), and the postoperative (D1) serum concentrations of GDF-15 in patients with SNVP-OD or SNVP-ODT increased significantly compared to D0 ($P<0.05$). The preoperative (D0) serum concentrations of GDF-15 in patients with MNVP-OD ($n=35$) and MNVP-ODT ($n=35$) were similar ($P>0.05$), and the postoperative (D1) serum concentrations of GDF-15 in patients with MNVP-OD or MNVP-ODT increased significantly compared to D0 ($P<0.05$)

during the 0–6 h interval in the SNVP group, there were fewer vomiting cases in the SNVP-ODT group than in the SNVP-OD group, which means that the ODT was also comparatively effective for PONV symptom controlling within 6 h after the operation. However, other potentially effective regimens require further investigation to reduce the incidence of nausea. Due to the lower incidence of PONV in patients in the MNVP group, it is reasonable that TEAS did not show a significant effect on PONV treatment 6–24 h after operation. Therefore, TEAS could be an effective method to control PONV in SNVP-risk patients.

This study tested the GDF-15 serum concentrations to verify whether it is a genetic factor predisposing patients with a history of SNVP to PONV. GDF-15 is a genetic factor that correlates with the development of emesis [17], nausea and vomiting during pregnancy [25], and chemotherapy-induced nausea and anorexia [26]. Serum levels of mature GDF15 dramatically increase during pregnancy and in many diseases, including cardiovascular disease, cancer, tissue damage, and liver and kidney diseases [27]. In this study, GDF-15 levels were significantly elevated after surgical procedures, which corroborated the increase in GDF-15 levels during tissue damage. GDF-15 is associated with poor prognosis in cancer patients [28, 29]. In contrast, this study found that higher GDF-15 levels are associated with more aggressive surgical procedures. Circulating GDF-15 levels are higher in cancer patients receiving chemotherapy [30], corroborating our findings that GDF-15 levels in patients receiving neoadjuvant chemotherapy were significantly higher than in those without neo-adjuvant chemotherapy. Meanwhile, it has been confirmed that there is an association between maternal GDF15 levels and HG and levels of nausea and vomiting



during pregnancy [16]. However, whether GDF-15 levels are associated with PONV has not been explored. Fortunately, this study helped establish that higher levels of

GDF-15 were associated with both SNVP and PONV and that GDF-15 might be the underlying genetic factor that predisposes patients with a history of SNVP to PONV (Fig. 3a and b).

This study has several limitations. First, plasma GDF-15 concentrations were only measured on D0 and D1, and shorter or more extended periods might lead to the discovery of more information. Second, owing to unpredictable sentinel lymph node involvement, the number of different surgical procedures was not balanced between the different treatment subgroups. The number of patients who underwent total mastectomy and axillary dissection was significantly higher in the SNVP-OD group than in the SNVP-ODT group. At the same time, the number of cases receiving segmental mastectomy and axillary dissection was not well balanced in the MNVP-ODT and MNVP-OD groups. Although the mean surgical time and opioid consumption of the different treatment subgroups were similar, it would be better if the number of different surgical types was balanced.

In conclusion, our study confirmed that women with a history of SNVP had a significantly higher incidence of PONV and severe PONV after breast cancer surgery than women without SNVP. TEAS is an effective treatment for PONV in patients with SNVP. A higher circulating level of GDF-15 was associated with SNVP history and a higher risk of PONV, establishing GDF-15 as a genetic factor that might predispose patients with a history of SNVP to PONV.

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Author contributions WBN, CHQ and ZH contributed to study design, data collection and analysis, drafting of manuscript. WSJ, HH, GJ, XMY, YJ and ZX contributed to data collection and interpretation, and revised the manuscript.

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Availability of data and materials All the data used and analyzed are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest All the authors declare that they have no conflict of interest.

Ethics approval This article does not contain any studies with animals performed by any of the authors. All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

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